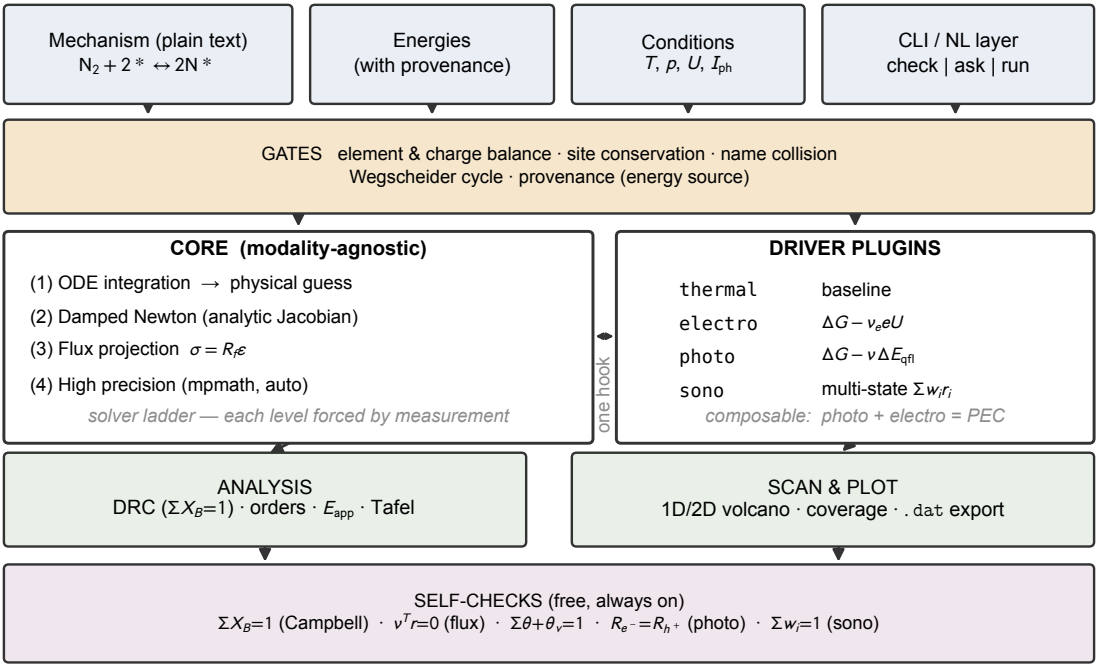

Mechanism (plain text)
$N_2 + 2* \leftrightarrow 2N*$
Energies
(with provenance)
Conditions
T, p, U, I_{ph}
CLI / NL layer
check | ask | run
GATES element & charge balance · site conservation · name collision
Wegscheider cycle · provenance (energy source)
CORE (modality-agnostic)
(1) ODE integration → physical guess
(2) Damped Newton (analytic Jacobian)
(3) Flux projection $\sigma = R_f \varepsilon$
(4) High precision (mpmath, auto)
solver ladder — each level forced by measurement
one hook
DRIVER PLUGINS
thermal baseline
electro $\Delta G - \nu_e eU$
photo $\Delta G - \nu \Delta E_{qfl}$
sono multi-state $\Sigma w_i r_i$
composable: photo + electro = PEC
ANALYSIS
DRC (ΣX_B=1) · orders · E_{app} · Tafel
SCAN & PLOT
1D/2D volcano · coverage · .dat export
SELF-CHECKS (free, always on)
ΣX_B=1 (Campbell) · $\nu^T r$=0 (flux) · $\Sigma\theta+\theta_v$=1 · $R_{e^-} = R_{h^+}$ (photo) · Σw_i=1 (sono)